

WASP-43b

R-2573

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_140308 · created 2026-08-27T00:08:06+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2456724.67544 +/- 0.00073 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.198 +/- 0.057
Transit depth (Rp/Rs)^2	3.93 +/- 2.24 [%]
Orbital Inclination (inc)	82.94 +/- 2.89
Airmass coefficient 1 (a1)	1.11 +/- 0.55
Airmass coefficient 2 (a2)	-0.04 +/- 0.19
Scatter in the residuals of the lightcurve fit is	52.79 %
Best Comparison Star	None
Optimal Aperture	2.76
Optimal Annulus	5.75
Transit Duration (day)	0.053 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.198 $\pm$ 0.057 vs 0.1594 (deviation 0.68 σ)
Duration fit vs published	0.053 d vs 0.0512 d (deviation 0.16 σ)
Mid-time O-C	-0.2 min
Depth SNR	3.5
Residual scatter	52.79 %

Light curve

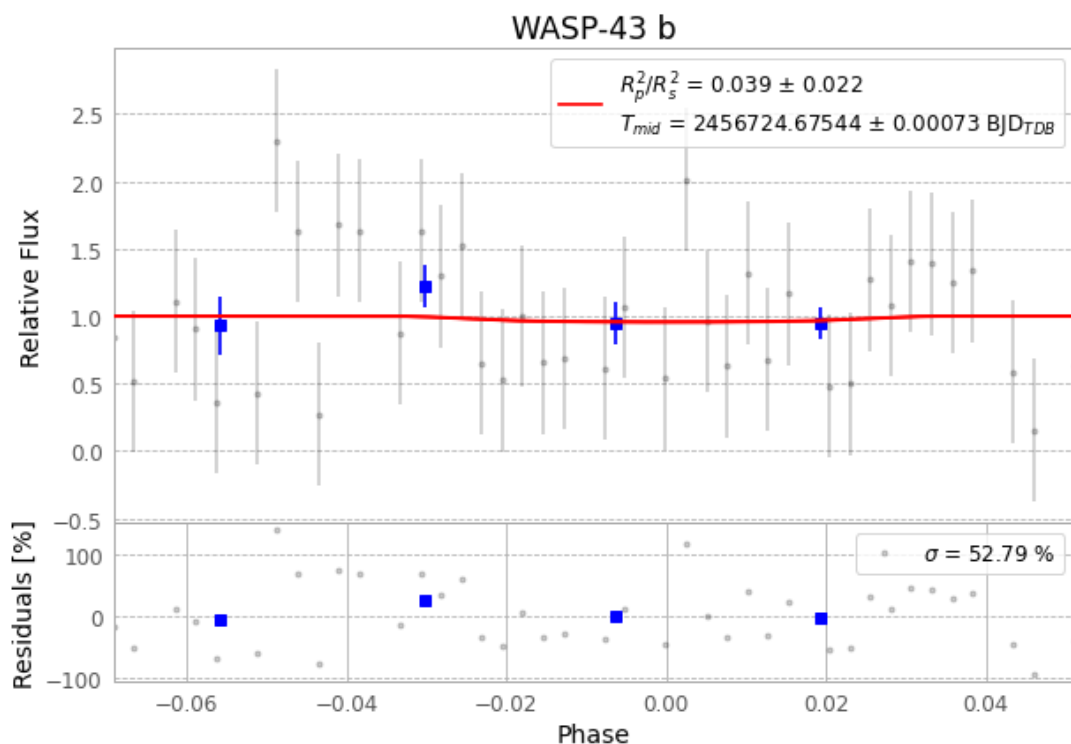


Figure 1 — FinalLightCurve_WASP-43 b_2014-03-07.png (SHA-256 4004fa9220d3d3f8...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [496, 147], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 958550eb6ae2e4b9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), WASP-43140308023912.FITS → WASP-43140308054812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-140308.log (860ced3efe7b...), FinalLightCurve_WASP-43 b_2014-03-07.png (4004fa9220d3...), FinalLightCurve_WASP-43 b_2014-03-07.csv (503a6dbc5dd7...)

Compute resources

Wall time 128.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-27 00:08Z.